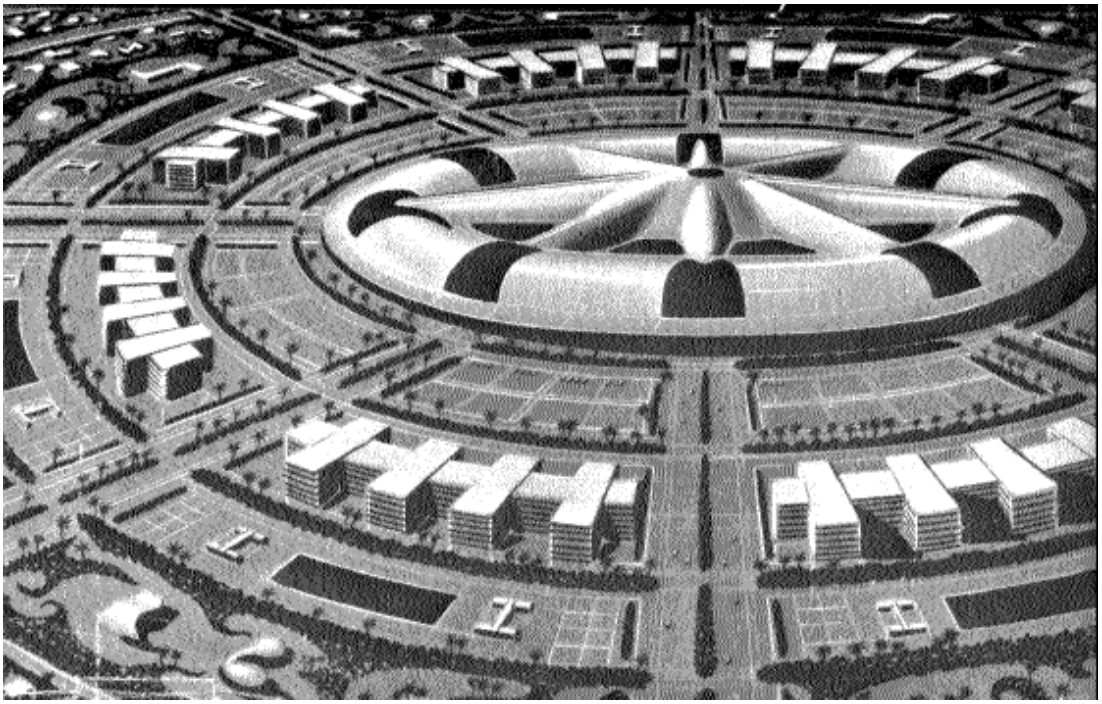
Looking Forward

KENNETH S. KEYES, JR. and JACQUE FRESCO

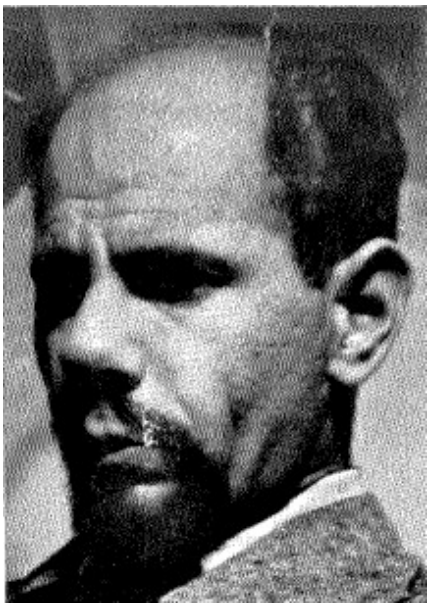


Dedicated to the youth of the world who must meet the challenge of the twenty-first century.

About the Authors



Kenneth S. Keyes, Jr., is a multi-faceted man. He attended Duke University and obtained a BA from, the University of Miami, majoring in psychology and minoring in music. His interests include art, symphonic music, yachting-he lives aboard a 71-foot yacht named Caprice and writing-he is the author of *How to Develop Your Thinking Ability* and *How to Live Longer-Stronger-Slimmer*. He has taught in the Evening Division of the University of Miami.



Jacque Fresco worked as an industrial designer for thirty years, designing all types of equipment from prefabricated houses to automobiles, electronic and medical equipment, human factors systems, and hundreds of commercial products and inventions. He has designed and patented such varying items as a radical aircraft wing structure patented by the USAAF and three-dimensional motion pictures not requiring the "use of viewers. Numerous articles and photographs have been published about his work in many magazines and newspapers. He has served as technical advisor in a number of motion pictures, including one of the first on space stations and a journey to the moon called *Project Moon Base*. He lives in Miami.

LOOKING FORWARD

by Kenneth S. Keyes, Jr. and Jacques Fresco

Looking forward is an imaginative and fascinating book in which the authors take you on a journey into the culture and technology of the twenty-first century. After an introductory section that discusses the "Things that Shape Your Future." you will explore the whys and wherefores of the unfamiliar, alarming, but exciting world of a hundred years from now.

You will see this society through the eyes of Scott and Hella, a couple of the next century. Their living quarters are equipped with a cybernator. a seemingly magical computer device, but one that is based on scientific principles now known. It regulates sleeping hours, communications throughout the world, an incredible underwater living complex, and even the daily caloric intake of the "young" couple. (They are in their forties but can expect to live 200 years.)

The world that Scott and Hella live in is a world that has achieved full weather control, has developed a finger-sized computer that is implanted in the brain of every baby at birth (and the babies are scientifically incubated the women of the twenty-first century need not go through the pains of childbirth), and that has perfected genetic manipulation that allows the human race to be improved by means of science.

Economically, the world is Utopian by our standards. Jobs, wages, and money have long since been phased out. Nothing has a price tag, and personal possessions are not needed. Nationalism has been surpassed, and total disarmament has been achieved; educational technology has made schools and teachers obsolete. The children learn by doing, and are independent in this friendly world by the time they are five.

The chief source of this greater society is the Correlation Center, "Corcen," a gigantic complex of computers that serves but never enslaves mankind. Corcen regulates production, communication, transportation and all other burdensome and monotonous tasks of the past. This frees men and women to achieve creative challenging experiences rather than empty lives of meaningless leisure.

Obviously this book is speculative, but it is soundly based upon scientific developments that are now known. And as the authors state:
"You will understand this book best if you are one who sees today only as a stepping stone between yesterday and tomorrow.

You will need a sensitivity to the injustices, lost opportunities for happiness, and searing conflicts that characterize our twentieth-century civilization. If your mind can weigh new ideas and evaluate them with insight, this book is for you.
"We have no crystal ball. ... We want you to feed our ideas into your own computer, so that you can find even better ideas that may play a part in molding the future of our civilization."

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HOW TO DEVELOP YOUR THINKING ABILITY
HOW TO LIVE LONGER—STRONGER—SLIMMER

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Acknowledgments

The authors are indebted to countless people for the ideas and encouragement that have made this book possible. Most of those who reviewed the manuscript felt that these enormous changes in man and his environment might happen in 1,000 years but not in the next century as we suggest. The authors, however, have wondered whether the future society they describe may be partially in existence by the time the book is published. We see so many of our predictions of things to come being discussed, developed and tried that we suspect we have been too conservative in estimating the time.

The following kind friends have read the manuscript and offered excellent suggestions—some of which were used: Anne Ammirati, John Bethea, Louise Boches, Janice Burr, Charles Kimball, Shirley Lewis, William A. McCall, Gretchen McCall, Graham Miller, Joe Prospero, Charles Ray, Christie Ray, Arden Richards, Velma Richards, Marjofie Sherrill, and Anitra Thor- haug. We are indebted to Herbert Wallach, Jr., for suggesting our title—Looking Forward. Bonita Bennett listened to these "far-out" ideas, typed them studiously, and still had the stamina to assist greatly in editing and revising. Shirley Rosichan offered many excellent editorial suggestions. Others who helped with various phases of the work were Marty Costello, Karen Brandt, and Stephanie Brovold. Frank Seldon and Carl Green tirelessly assisted in reproducing copies of the manuscript. Also thanks to Iván García who helped digitalizing the paperbook trough OCR. Our thanks are also due to the authors and publishers who have kindly given permission to quote from their works.

Kenneth S. Keyes, Jr.
Jacque Fresco

Miami, Florida

PART I
THINGS THAT SHAPE OUR FUTURE

1.The Leap from the Jungle

The lives of most men and women are blighted by problems they cannot solve. And people usually blame themselves, or they blame "fate" whatever that is. However, when two cars collide at an intersection, should we, as students of society, concentrate our attention on the individual blame of the drivers, on "fate" or on the way transportation is engineered so that it permits collisions?

If you believe that cars and roads should be designed so that it is almost impossible for people to lose their lives through collisions, this book is for you. If you believe that the mind is capable of gradually applying the method of patient, scientific investigation to find out how to rearrange the structure of our society to give each individual a greater opportunity for self-realization and happiness while he is on earth, we welcome and need your help.

If you believe it's about time for the human race to stop spinning its wheels, then let's get going!

But this book on the future of our civilization is not for everyone. Few will be able to read it without forming an opinion before they see the picture as a whole. To enjoy this book you will have to blend open-mindedness with critical skepticism. It is hard enough to face the problems of our own time. And it is many times more difficult to understand a projection of fantastic and shocking changes that may occur over the next hundred years!

Suppose an intelligent man in New York City around 1860 had sat down one evening with a book predicting life a century later.

He would have refused to believe that almost everyone in 1960 would be able to own a horseless carriage that could whisk about at 60 or more miles per hour. With his Victorian attitudes he would have been deeply shocked by the brief bikinis. In 1860, not even a "woman of ill repute" would have appeared so undressed in public. He would have smiled smugly at the ridiculous prediction that man-made flying machines would travel faster than the speed of sound. The thought of sending pictures and sounds through miles of air would have seemed impossible to a sensible person in 1860. It would have been incredible to him that the art of war would progress to the point where one small bomb would destroy a city. Our Victorian would have been alarmed that a part of his wages could be withheld to provide for retirement. At this point, let us leave this gentleman of the last century muttering to himself about loss of freedom and the world's moving too fast.

Are we more flexible—more farsighted—today? We will need to become experts at changing our minds. The differences between the nineteenth and twentieth centuries will probably be small when compared with the accelerating pace of the next century.

You will understand this book best if you see today as only a stepping stone between yesterday and tomorrow. You will need a sensitivity to the injustices, lost opportunities for happiness, and searing conflicts that characterize our twentieth-century civilization. If your mind can weigh new ideas and evaluate them with insight, this book is for you.

We have no crystal ball that gives an accurate picture of the twenty-first century. We want you to feed our ideas into your own computer. Perhaps you may find even better ideas that may play a part in molding the future of our civilization. In the next six chapters we'll explore the "why" and "wherefore" of the unfamiliar, alarming, unbelievable, wonderful, and exciting picture we will paint of the twenty-first century. Then we'll join Scott and Hella, who live in the next century. We'll experience with them the new dimensions of life in the changed world of the future.

The Long Journey

To understand the probable courses of man's future development, let's spend a few minutes looking into his past. The world came into existence around four and one-half billion years ago, and all sorts of weird fishes and monstrous dinosaurs got into the act before we did. Millions of years ago our ancestors were little apelike fellows that spent most of their time in trees. Then some of these little beasts began to do things that were to make a lot of difference to you and me. They quit jumping from branch to branch like the squirrels and, instead, began swinging from limb to limb somewhat like the man on the flying trapeze. This led to some important changes from head to toe. The arms, which previously moved in a restricted arc, developed free rotation.

This makes it possible for a baseball pitcher to whirl his arms around and zing one over home plate. The intestinal organs, which had been slung from the backbone, as in a dog, were now supported by the pelvis, which became somewhat bowl-shaped.

The front feet didn't have to support the weight of the body anymore, and they developed into a bunch of skyhooks that we call fingers. Since animals that misjudged distances when swinging from branch to branch left fewer offspring, we are blessed today with excellent stereoscopic vision and neuro-muscular co-ordination. We owe a large part of what we are today to our swinging primate ancestors.

Man has made three big steps away from his animal cousins. The first cultural jump occurred when he began to use fire, tools, and language. Although men of our own species, Homo sapiens, have been here about 50,000 years, radioactive carbon datings show that our ancestors were using tools and fire as much as 600,000 years ago.

The beginnings of language probably occurred several hundred thousand years ago. This was a tremendous leap forward. The development of language may have played a part in helping us be as intelligent as we are today. Suppose someone had yelled, "Watch out for the tiger in the tree behind you!" The more intelligent ones would have got the message fast. They probably charmed the greater number of females that evening and, thus, left more offspring.

The second big cultural jump taken by our ancestors occurred about 7,500 years ago. This was the discovery of how to raise food. The development of agriculture and animal-raising made it possible for us to live in crowded nests known as cities. When man had to gather his food in the form of fleeing animals and random plants, it took a lot of land to support a small group.

For example, there were probably not over 100,000 people on the entire continent of Europe before they learned to raise food.

A good year might have increased the birth rate. But they would soon have been killed off by famine and disease if there were even one rough year when the game and plants were less available.

When man began to raise food, he could stay in one place instead of roaming all over the countryside. Socially and technologically, lots of things began to happen for the first time. He began to accumulate bric-a-brac. The wheel was developed. He learned how to heat metals to make them soft so that they could be poured or beaten into useful shapes. He developed the plow for working the land and the loom for weaving cloth. Social patterns that were needed in city life were developed. He amplified political structures and created armies equipped with death-dealing instruments. Within a thousand years after our ancestors acquired the know-how for raising food, the cultural patterns of city life, politics, business, and technology were invented. Since then they have continued with very little change until recently.

Many anthropologists consider the city as our most fundamental social invention. The first cities evolved in southwestern Asia, and the pattern of the city was well worked out in Mesopotamia by 4,500 to 4,000 B.C. Cities did not appear in China until around 2,000 B.C. Europe had to wait until the Greeks put together a few cities around 900 to 800 B.C. The city did not appear in Scandinavia until after 1,000 A.D.

Writing developed almost simultaneously, about 5,000 to 6,000 years ago, in Egypt, Mesopotamia, and the Indus Valley. The Chinese invented their cumbersome symbols 2,000 years later. When a man's thoughts were written, they could speak out after his death. The human race began to accumulate information that permitted the building of a modern civilization with a complex value system.

Whereas learning was previously a monopoly of a privileged class, the development of the alphabet eventually made it possible for everyone to acquire an education. All alphabets now in use seem to stem from a single point of origin in the Sinai Peninsula where the Egyptians were busy mining. Most of the time the Egyptians used criminals and prisoners of war for this type of work. Occasionally, some wandering Semites who needed food came to work for them. Since they were pretty smart fellows, the sheiks of the Semites were appointed foremen of the mines.

As part of their job they drew up reports on how much metal was mined and how much money was earned by the various men.

From the complicated Egyptian system of writing, these foremen abstracted the simple symbols representing single sounds in the Egyptian system. We are indebted to them for giving us the first alphabet.

We were doing business as usual by at least 3,000 B.C. Standard weights and measures were in use in Sumeria. Barley was one of the earliest mediums of exchange. A measure for barley was available for

public use in the market place. If a farmer had felt he was being gypped, he could have walked over and measured it out. By 3,000 B.C. convenient metal coins began to be used in place of the more awkward sacks of barley, and a shortage of money became a continual problem for the government. The Code of Hammurabi set the value ratio of barley and metal. This code provided serious penalties for anyone caught cheating. To make sure this money system would work, a merchant could have been put to death if he had refused to accept either barley or coins in payment for his merchandise.

There are cuneiform records of loans with interest that ran as high as 300 per cent when an individual could not offer security.

Even the lowest rate was 25 per cent per year. The Sumerians worked out business methods such as stock companies and corporations. Many tablets recording their business deals and their private correspondence have survived. One tablet dated before 2000 B.C. describes the complaint of an old man about the degeneracy of the younger generation. Another tablet is from a boy at school telling his parents about the "lousy food."

The late Ralph Linton , a noted American anthropologist, wrote:

> *Many of the economic and social patterns which still operate in modern Western society can be traced to this region. It has been said that if George Washington had been transported back to the court of Hammurabi of Babylon, about 2067-2025 B.C., he would have felt vastly more at home there than he would in the modern capital city which bears his name. Apart from language difficulties, he would have encountered very few things in Hammurabi's empire which were not familiar and understandable, while in Washington he would have been baffled and confused by the tremendous technological changes . . .*
> *and the fumbling efforts which our society is making to bring the other aspects of its culture into adjustment with these.**
>
> ** (The Tree of Culture (New York: Alfred A. Knopf,1959), p. 298.)*

The third enormous step in the development of human culture took place when we began to supplement the puny muscles of human beings and animals with other sources of power. The use of sails on boats eventually replaced galley slaves. Water power was used to carry things downstream. The water wheel furnished power to run mills. Later, steam power replaced even more human muscles. During the last hundred years we learned to manufacture large amounts of electrical energy. Then things began to hum because this made energy or power available at any point to which we could run a wire or lug a generator.

From a modern point of view one of the most useful measures of the development of a civilization is the amount of available energy per person. To a large extent the degree of physical comfort that you enjoy today is correlated with the energy that is at your disposal.

The application of the scientific method of thinking has made it possible to develop almost unlimited amounts of energy. This energy may be in the form of electricity which will run a constellation of labor-saving and life-enriching appliances and instruments. It may be in the form of coal, gasoline, oil, or nuclear power. It may be in chemical form, such as in an automobile battery or a flashlight cell. Imagine

the almost complete paralysis that would occur if your electricity and gasoline supply were cut off, and you had to use your own muscles in place of the complex of machines upon which you now rely.

Our Rapidly Evolving Civilization

We are today but a few steps from the jungle. While we've been trying to get away from the animal patterns of the jungle for a little over a half-million years, we really got moving only a few thousand years ago with the development of cities and the invention of writing. It has just been in the last century that we have started the large scale use of non-muscular sources of energy and power. Most factory workers today are laboring in industries that were not even in existence in 1900. Although our world may in some ways appear stable to us, we are in a furious transitional phase in which changes are occurring at the fastest rate in history.

Today we are at the beginning of this third phase of the development of our civilization. Fantastic developments lie ahead.

If life at times seems bewildering, if you feel pulled in many directions, if you find that no matter what you do, you still have sticky problems, if you find that our economic, political, and social ways of doing things sometimes create more difficulties than they solve, then you are simply playing your part in suffering through the present transitional phase of our civilization.

Much of your life is patterned along the lines used in western Asia several thousand years ago. Yet, some of the conditions to which you are trying to adjust have come out of the laboratory in the last few decades. If the day-by-day pattern of your personal, business, and social life is something less than serene, you've been caught in the wringer of change, and you've got lots of company.

2.The Confusion of Our Times

The habit patterns of men and women that may have been appropriate several thousand years ago cannot be made to yield maximum happiness in the changed world of today—to say nothing of the future civilization toward which we are rapidly evolving. This chapter will briefly catalogue some of the things that keep us frustrated, insecure, and jumpy. As might be expected in a time of rapid transition, few of the basic needs of men and women are now met in a satisfying way. We hope you will bear with us as we haul out some of the dirt that usually stays lumpily under the rug.

Among the hangovers from the past, we might list the grisly pattern of war. Back in Mesopotamia a war might have chewed up a few thousand people. The First World War killed approximately ten million people of whom 5 per cent were civilians. But that's only the beginning. The Second World War rolled up a death toll five times as large—approximately fifty million. About 50 per cent were civilians.

Suppose there were no police and no laws in your city. Who would be safe? Criminals might like it. But not you and your family. Similarly, the lack of a respected and enforced international law between nations endangers everyone on earth today. It's like living in a jungle.

General Eisenhower has summed up the tragic effects of the custom of war as a method of settling disputes between nations:

> *... a life of perpetual fear and tension; a burden of arms draining the wealth and the labor of all peoples; a wasting of strength that defies the American system or the Soviet system or any system to achieve true abundance and happiness for the peoples of this earth . . .*
> *Every gun that is made, every warship launched, every rocket fired signifies, in the final sense, a theft from those who hunger and are not fed, those who are cold and are not clothed.*
> *This world in arms is not spending money alone.*
> *It is spending the sweat of its laborers, the genius of its scientists, the hopes of its children.*
> *The cost of one modern heavy bomber is this:*
>
> *A modern brick school in more than 30 cities.*
> *It is two electric power plants, each serving a town of 60,000 population.*
> *It is two fine, fully equipped hospitals.*
> *It is some 50 miles of concrete highway.*
> *We pay for a single fighter plane with a half a million bushels of wheat.*
> *We pay for a single destroyer with new homes that could have housed more than 8,000 people . . .*
>
> *This is not a way of life at all, in any true sense. Under the cloud of threatening war, it is humanity hanging from a cross of iron.*

When Bertrand Russell, world famous philosopher, participated at the age of ninety in a big peace demonstration in London in 1962, he was arrested and sent to jail for a week! It seemed that he wanted to change some of our habit patterns. He wished to end nuclear bomb testing, get rid of all nuclear weapons, and abolish the institution of organized murder that we call "war".

Economic Insecurity

In spite of our program of so-called "social security," both personal and economic insecurity are usual in our time. Few people can be sure of continued employment. In many cases one can not be sure that the business that employs him will be in existence a year from now. Often, one can not be sure that the type of work for which he has been trained will be in demand next year. Automation is eliminating hundreds of thousands of jobs each year in the United States.

We are quite ingenious in building factories that can turn out enormous quantities of television sets, automobiles, refrigerators, toothbrushes, packaged foods, etc. But we wouldn't dare let these factories run full time because they would produce more goods than we could sell. So we have idle factories some of

the time with large numbers of people in need all of the time. Paradoxically, it seems that only when we're fighting a war are we able to keep our industrial machines operating at full capacity. Although much of our population can use better clothing, better housing, and better food, our techniques in distributing the bounty of mass production are bogged down by economic diseases that go under the names of "overpopulation," "unemployment," and "lack of purchasing power."

Dr. Ralph Linton has observed:

> *As the disharmonies within the culture become increasingly pronounced, more and more of the society's energy and resources have to be expended on makeshift adjustments until the period of rapid change gradually grinds to a halt. Our own society would seem to be in such a period at the present time. Its tremendous and still accelerating development of science and technology has not been accompanied by an equal development in social, economic, and political patterns . . . We are now . . . only beginning to explore the potentialities which it offers for developments in our culture outside technology, particularly in the social, political, and economic fields. It is safe to predict that even two or three centuries from now, such social inventions as modern-type Capitalism, Fascism, and Communism will be regarded as primitive experiments directed toward the adjustment of modern society to modern technology.**
>
> ** Linton, The Tree of Culture (New York: Alfred A. Knopf, 1959), pp.47-8.*

Our Polluted World

Although this is the only world we have, we're certainly not taking good care of it. We've already exploded enough atomic bombs to contaminate the atmosphere with strontium 90 and other radioactive elements. According to Hathaway and Leverton of the U.S. Department of Agriculture:

> *Strontium 90 . . . may become a health problem. Its radioactivity is slow to disappear, and its accumulation in the body could be dangerous. . . . Strontium 90 was first detected in animal bones, dairy products, and soil in 1953. It now occurs in all human beings regardless of their age or where they live.**
>
> ** Milicent L. Hathaway and Ruth M. Leverton, "Calcium and Phosphorous," Food, The Yearbook o/ Agriculture, 1959 (United States Department of Agriculture, Washington, D.C., 1959), p. 117.*

Atomic bombs are not the only means we use to pollute the air we breathe. The exhausts of automobiles, the excretions of industry, and the burning of coal, oil, and garbage are also busy contaminating the air. Today air pollution affects in some degree more than 7,000 urban areas inhabited by 115,000,000 Americans. When bituminous coal and low grade fuel oil are burned, sulphur dioxide gas and smoke are released. A five-day killer smog hit London in 1952 and killed 4,000 people. Air pollution has been linked with the common cold, asthma, pneumonia, tuberculosis, influenza, chronic bronchitis, pulmonary emphysema, and lung cancer.

Those who are not overly worried about human beings might well ponder other costs of air pollution. Sulphur dioxide has been shown to disintegrate nylon stockings. It reacts chemically with moisture to form sulphuric acid, which eats away roofs, eaves, downspouts, and other exposed metals. It even affects stone buildings and the proud statutes of military heroes in the parks.

Air and water are the most immediate physical necessities of life. Lest we neglect the latter, it may be pointed out that our largest river is well polluted with oils, phenols, ammonia and toxic metals, blood, refuse from hospitals and undertakers, and acids from mines. For example, in the St. Louis area there have been times when the chicken feathers, viscera, and offal collect in patches too thick to drive a motorboat through. The 300,000,000 gallons of sewage a day coming from St. Louis and East St. Louis contain 460 tons of solid garbage and 165 tons of ground garbage. In the part of the Mississippi from St. Louis to New Orleans, a mouthful of water contains half a million coliform bacilli, which come largely from untreated sewage. Nor does the United States have a monopoly on polluted water. Throughout the world about five million children die each year before their first birthday from diarrhea or dysentery through drinking unclean water.

Ephemeral Machines

The pleasure of living in the twentieth century is somewhat tarnished by the constant deterioration of the machines and gadg ets we use. They are, unfortunately, designed for a short life— both style-wise and function-wise. It is possible to build a washing machine or television set that would probably not need service during a ten-year period. It is possible to design an automobile that may remain trouble-free for a ten-year period and to give it an attractive design that would be appreciated even longer. Instead of designing for longevity and service, automobile designers rack their brains for ways to make next year's car so appealing that we'll trade in this year's car. Gerald Piel informs us:

> *According to the standard practice of our durable goods industries—always with the aim of perpetuating scarcity in the face of abundance—the automobile is designed for 1,000 hours of service, to be traded in at 40,000 miles or less.*

Can you think of any electrical or mechanical device in your home that will remain trouble-free for a long period? There is only one device in your home that is designed to last about twenty years. It is likely that it won't require service during this time. And you don't even own it! Although it is a complicated electronic instrument, you can drop it on the floor, and it probably won't break. Whether you use it once a day or continuously throughout the day, it does not seem to wear out. The reason it was designed to give at least two decades of trouble-free service is that it was not made to be sold. The company that rents it to you would have to repair it at their own expense if something were to go wrong. So they make sure that this complex device is engineered to high standards. They can't make money if it breaks down every few months. In case you haven't guessed the name of this one thing in your home that has been engineered to give you maximum utility and minimum maintenance, it is the telephone! It is designed for trouble-free use, not for cheap production at a profit. How wonderful it is to realize that there are ways to get off the treadmill of buy it, use it, junk it, and buy it again!

Standing Room Only

Overpopulation of the world is a twentieth-century problem. It took until about Ï800 for us to spawn a world population of one billion. Current predictions indicate that the present population of three and one-third billion will explode to seven billion by the end of this century and to sixteen billion by the year 2040 if the growth rate is not changed.

This enormous growth in population in many areas of the world is greatly exceeding the ability of most countries to provide food and decent living standards. The FAO World Food Survey of 1963 found that at least 60 per cent of the people in the under-developed areas were undernourished and that half of the world suffers from hunger, or malnutrition, or both. In Africa, Latin America, and the Far East, food production is growing only two-thirds as fast as the population. In the latter two areas the per capita food production is still below the levels attained twenty-five years ago ! Eugene R. Black, former president of the World Bank, has summed it up in this way:

> *I must be blunt. Population growth threatens to nullify all our effort to raise living standards in many of the poorer countries. We are coming to a situation in which the optimist will be the man who thinks present living standards can be maintained.*

Cultural Dilemmas

Most of the built-in dilemmas that face human beings in our fast-changing world pose problems for which the wisdom of the past offers no effective solutions. In so many ways we're like a man chased to the edge of a cliff by a roaring lion. If he jumps, he'll be hurt. And if he stays there, the lion's going to get him. For example, we have built-in sexual drives that become strong in the teen years. Marriage may seem to offer a solution. Yet, the marriage counselors advise us that early marriages have a greater rate of divorce—that a person needs to experience life and achieve a degree of maturity before choosing a life partner.The individual suffers no matter how he tries to solve the problem. If a person has an active sex life before marriage, the mores of our tribe may load him with feelings of guilt. If an unmarried person denies his need for sexual expression, he may enjoy a culturally-clean conscience, but he will be fighting a constantly stimulated, deep need that has been structured into his body. Guilt feelings may still arise from vivid sexual fantasies, erotic dreams, or masturbation. No matter what choice a person makes in this situation, it is usually accompanied by conflict and doubt. Not only in matters of sex, but in most business, personal, and social matters we are confronted with countless dilemmas. Our present folkways make it difficult to achieve effective solutions that deeply contribute to human dignity and happiness.

On the Homefront

"The mass of men," said Thoreau, "lead lives of quiet desperation."
The lives of most people in our present civilization fall far short of fulfilling levels of serenity and happiness. Ann Landers, a newspaper columnist, who deals with personal problems received this poignant letter:

Dear Ann Landers :
How do you feel after reading a couple of hundred letters? Disenchanted, I'll bet.
When I look back at my own life, its problems and its failures, I wonder what is it all about?
Then I look at my children and what has happened to their lives. Dear God, I tried. He knows I tried. But where did I fail? I must have failed. I am their mother.
The children went to church and Sunday School. They earned Bibles for perfect attendance and they had love. But here is the record :
One married while in the Air Force. Five years later—debt, drinking, divorce. Two children, two unhappy pawns. Remarriage out of the church. More debts.
Our daughter, desperately in love in high school, married to a fine but sick man. Willing but unable to work. Debts, then death.
A new love? She thought so, but her second marriage was a poor one. Where will it end? Heaven only knows. . . .

The lives of not-so-quiet desperation of many people are reflected in another letter received by Ann Landers.

Dear Ann:

Our five children are in bed and I am looking at a huge basket of clothes that I should be ironing, but I'm writing this letter to you instead.
I'm so exhausted if I walked past the bed and looked at it I'd fall asleep standing up.

My husband is a wonderful person and a terrific father. He doesn't drink or gamble and wouldn't think of spending a dime on himself. He always puts me and the children first. He works hard at his job, but every week they take something out of his paycheck. I don't think we'll ever be a dollar ahead.In 18 years this house will be paid for, and then it will probably fall apart.
The kids, God love them, are wonderful. They help me by doing for themselves and by just being good. Never do I hear a complaint because they have to wear their cousin's hand-me-downs or because there is no money for treats or Scout uniforms.
If I could get my hair done in a beauty shop and eat dinner out once in a while I'd think I was in heaven. Is this what life was meant to be?

Tired

*Ann Landers began her reply, "Only if you're lucky. . . ." **

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The Quandary of Women

It is quite possible that most women in our rapidly changing civilization have a rougher time of it than men. A woman who lived on a farm two centuries ago was deeply needed and felt secure. She and her home were the center of vocation, recreation, and education. Although she worked hard, she was psychlogically secure in her own feeling of worth. She was confident of the great need which she supplied in the lives of her husband and children. Today, household gadgets have relieved woman of some of the work, but the secure emotional foundations of her life have been largely swept away. The home is no longer the place where the family makes its living. Factories and offices beckon the father to a world of business not shared by the rest of his family. Schools grab up the children and take over the responsibility for educating them—frequently, in a way that is quite different from the training of the parents. Although television has added to the recreational aspect of modern homes, most of the really exciting things happen away from home. Automobiles scatter the family in all directions, and the home is often used primarily as a hotel in which to eat and sleep.

The modern housewife is expected to be a fascinating and energetic companion to her husband. She must meet the endless needs of her children. And at the same time she must operate the household, including food-gathering at the grocery store. In addition to these three full-time occupations which stretch her out pretty thin, she should find time to develop her own mind and body, including frequent trips to the beauty salon. After a number of years of this tearing in every direction, most wives begin to feel that, "Life is passing me by." They begin to question their self-worth. They realize that they are needed less and less by their husbands and children. Unhappiness, divorce, suicide bitterness, and blighted personalities are often the consequences of our rapidly changing culture.

In the United States there are currently around four million divorces a year that affect about a third of a million children mostly under ten years of age. Divorce statistics themselves are not important, but the painful bickering, searing accusations, and damaged egos that lead to a divorce greatly affect the total sum of human happiness.

When a woman tries to live a fuller life in the world outside her home, the path is not smooth. Dr. Ruth B. Kundsin, a prominent Harvard bacteriologist, said:

> *It is my contention that women in the United States have been victims of both prejudice and discrimination. . . If she has a sprinkling of Harlowian traits, her male colleagues are flirtatious. If she looks like a benevolent moose, they are merciless and their appraisal of her appearance takes precedence over what she has to say.*
>
> *Dr. Kundsin would have us consider the enjoyment of living with a happy, independent woman who is utilizing her talents completely and arrives home with sparkling, challenging expertenees of her own to tell and share. . . . Couldn't it be that the love of such a woman is a wondrous, exciting experience? Or does the American male ego really need a female slave in residence?*

Just in case it's beginning to look as though only the women have a rough time of it, there's a tragedy in the monotonous, routine-filled hours by which most men earn their living. Spending most of the prime years of one's life fighting to earn an existence severely limits the self-realization and happiness of most men. The world is so full of exciting things to learn, of interesting places to travel, of countless creative hobbies, of body-building sports, and of lovely sunsets and natural beauty that one must wistfully sigh at our limited opportunity to experience the best things in life.

A Hornet's Nest of Problems

In this chapter we have briefly touched on some of the problems of our confused civilization that scream the loudest for solutions. It would be possible to write several books just describing the "hangovers" of our transitional civilization. Take a deep breath because we haven't covered even a hundredth of the problems of the present. Help us fill out the long, long list:

> *The failure to apply scientific methods to solve social problems; the thousands of insidious types of prejudice and prejudice-inspired violence and killing; the hollowness of so much of family and social life; a competitive life pattern that makes it difficult for us to give fellow humans the deep acceptance that their egos need so much; the failure of men to cooperate in joining hands over national boundaries to build a civilization in which all men may be happier, the failure to seek and utilize the abilities of geniuses, the shortage of good teachers; schools that indoctrinate in what to think instead of teaching how to think; the unfortunate attitude that education stops when "schooling" is over; the failure of 40 per cent of our young people to finish high school; the restlessness of our adolescent population; drug addiction; the problem of abortion; mental illness; the failure of prisons to rehabilitate; the growing crime rate; slums that mar our beauty and disgrace our humanity; a countryside made ugly with shoddy stores, screaming signs, and junkyards; the primitive automotive systems which kill over 40,000 people each year; the high rate of all kinds of accidents that kill one person in the United States every six minutes and injure someone every three seconds; the difficulty of consumers in getting reliable information on products because of advertising that exaggerates little differences; the competitive system in which one man's success in business may throw three into the tragedy of bankruptcy; unemployment; the insecurity of men and women in the forty to sixty-five age range who have difficulty in finding well-paying, dignified employment; price-fixing; industrial strikes; the squeeze on the small businessman; monopoly; the graft of politicians and the hypocrisy of lawmakers; high taxes; senior citizens, restless in retirement; unscientific tampering with our land, resulting in floods, dust storms, and the loss of valuable forests; the fertile acres wasted in the growth of tobacco that is detrimental to the health of mankind; water shortage that hampers agriculture and industry:*
> *the use of pesticides in ways that injure people and kill wildlife; agricultural practices and processing techniques that result in low quality foods; choosing foods by taste and habit instead of by vitamin-mineral-protein-unsaturated fat content; immature personalities reinforced by the deficiencies of television, radio, and motion pictures; the prevalence of inadequate values based on wealth and social prestige, which seldom bring happiness when achieved; and the list may go on and on and on.*

The purpose of this book is to show how the mind and heart of man can solve these apparently "unsolvable" problems. The readers of this book may be "pioneers" in a deeply significant way when the long-range story of civilization is written.

3.Predicting the Future

> *At every point in history [warns Dr. George Gallup] man has assumed that civilization has reached its zenith. He has smugly refused to place himself on a scale of time that reaches thousands and millions of years into the future as well as into the past. Looked at from the vantage point of 8,000 years hence—approximately the period of recorded history—man's progress up to the present time may appear far less impressive than it does today.**
>
> ** George Gallup, The Miracle Ahead (New York, Evanston, and London: Harper & Row, 1964), p. ix.*

We have been here such a short time that we could almost be called "newborn." If you were to use a twenty-four hour clock to represent the time since life began on earth, it would show that man has only been in existence since the last minute of the twenty-fourth hour; only during the last few seconds of the last minute has modern man begun to use scientific methods to lead him to the most effective ways to get things done. We are just now beginning to hit our stride. More new knowledge has been created during the twentieth century than in the previous billion years. Change is everywhere.

How does one go about predicting the most probable changes in man's future? One might at first think that scientists could give us worthwhile information on the future. They're busy hammering out the next step. They're running experiments to find out what works and what doesn't. They're patiently sifting facts and theories that form the stepping stones to the future. But a glance at the record shows that few scientists have been able to anticipate future developments correctly. They have often been woefully wrong in giving opinions on the probability of events even a decade in the future.

About eighty years ago Thomas Edison, after his brilliant success with the phonograph and carbon microphone, became interested in using electricity to make light. When the news of this got around, the securities of the gas companies began to drop. The British Parliament appointed a committee to investigate the possibility of developing an electric light. The concensus of the experts was that Edison's ideas were, "good enough for our transatlantic friends . . . but unworthy of the attention of practical or scientific men."

C. D. Darlington the brilliant English geneticist said:

> *It is no accident that bacteria were first seen under the microscope by a draper, . . . that oxygen was first isolated by a Unitarian minister, that the theory of infection was first established by a chemist, the theory of heredity by a man who was unfitted to be a university instructor in either Botany or Zoology.*